

# **Building Components and Services**

- Building components are the different parts of a building that provide strength, stability, protection, and functionality.
- Building services are systems installed in a building to ensure comfort, safety, and efficient operation.
- Building Components are : Substructure Components (Foundation, Plinth, DPC), Superstructure Components (Walls, Columns, Beams, Roof, Doors, Windows, Stairs)
- Building Services are : Water Supply System, Sanitation and Drainage System, Electrical Services, HVAC (Heating, Ventilation, and Air Conditioning), Fire Protection System, Communication Services, Security Systems)

## **Door:**

- A door is a hinged or otherwise movable barrier that allows entry into and exit from an enclosure.
- A framework of wood, steel, aluminum, glass or a combinations of these materials secured in an opening in a wall for the purpose of providing access to the users of the structure.

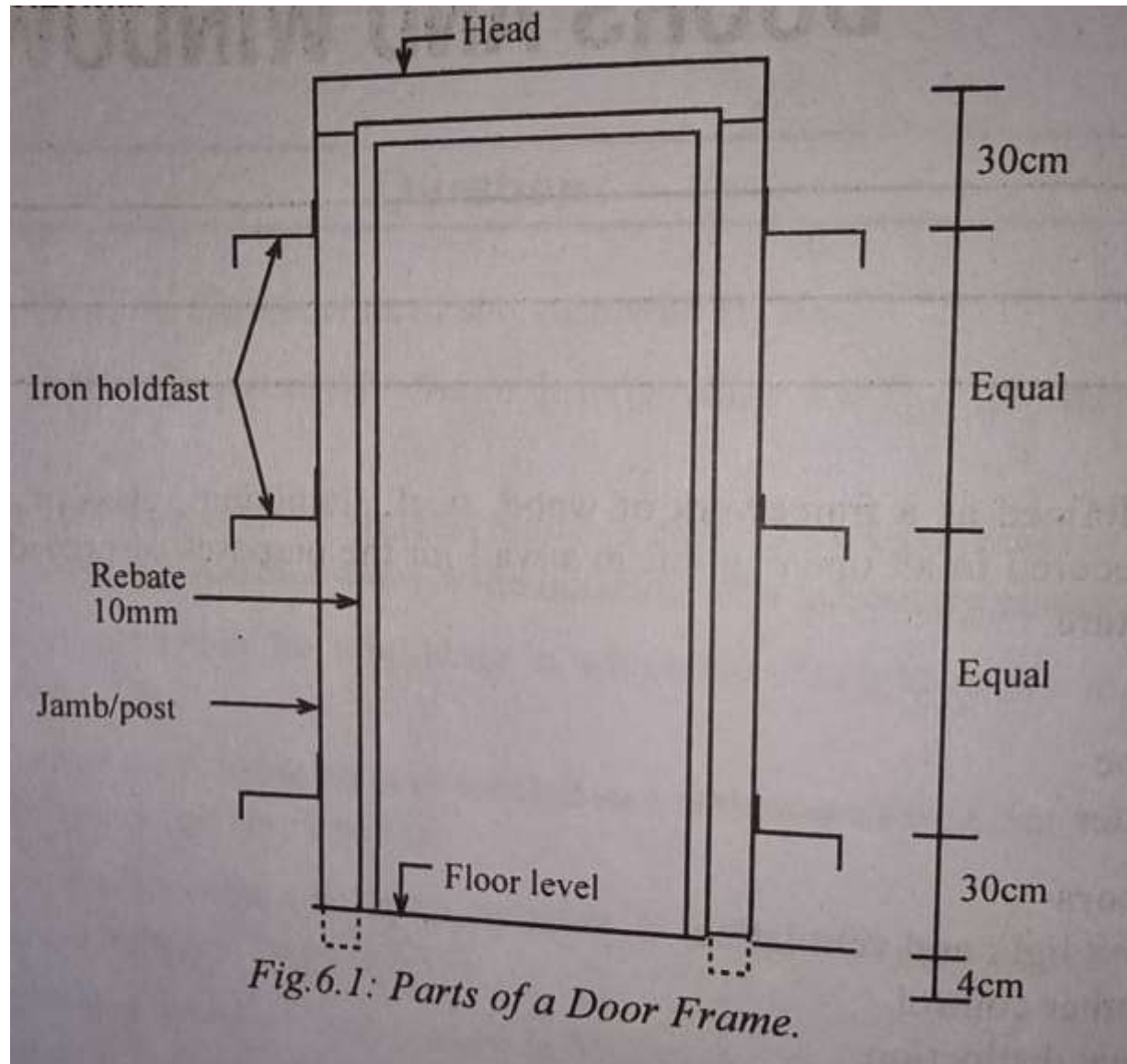
## **Function of the door:**

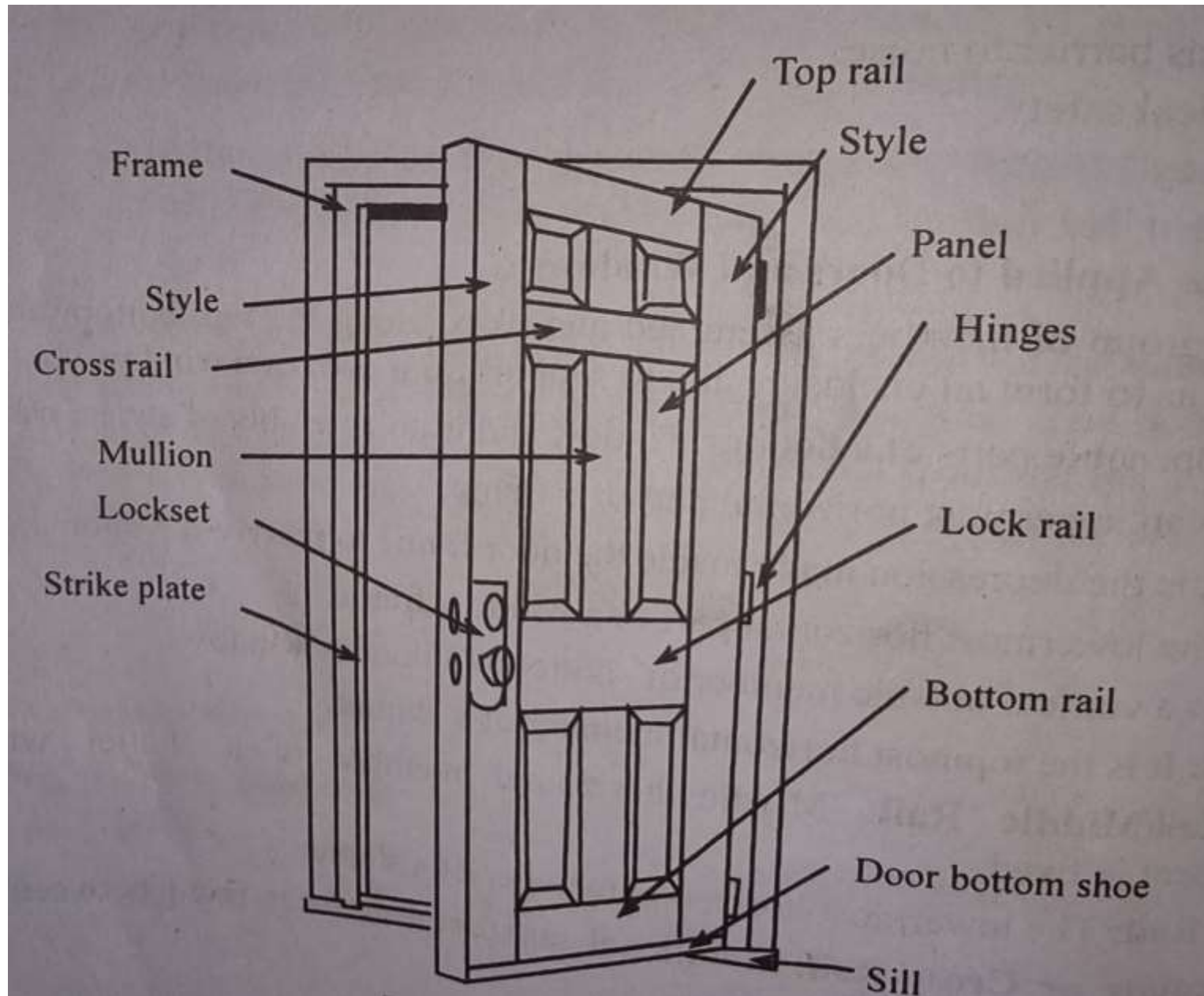
- Admit light and ventilation.
- Weather control.
- Visual distinction.
- Act as barrier to noise.
- Physical safety.
- Privacy

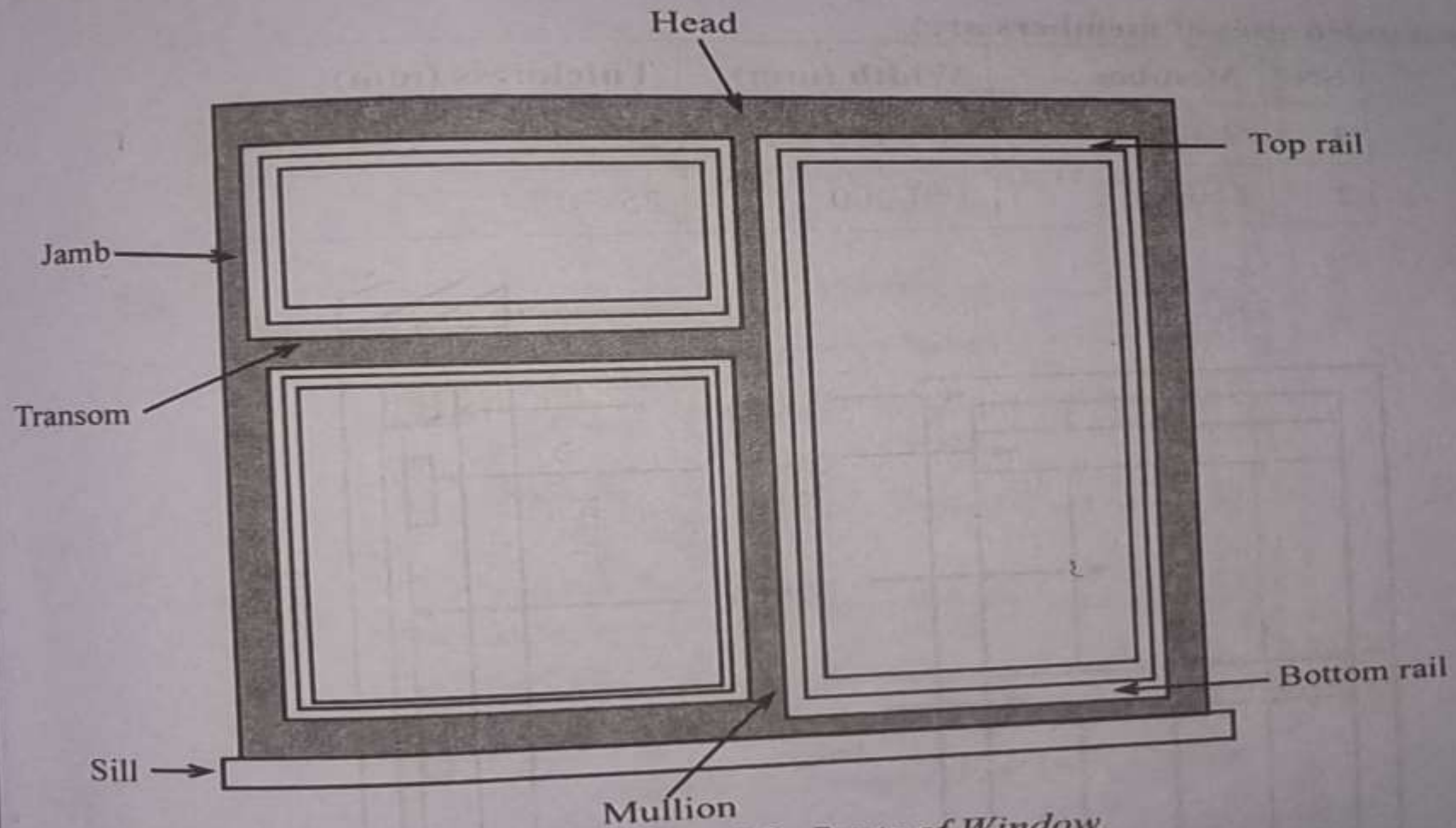
## **Technical terms applied to doors & windows:**

- **Frame-** A group of members assembled and place along the top, bottom & sides of an opening so as to form an enclose and to support for a door & window shutter.
- **Shutter-** Openable parts of door or window and is an assembly of styles, planes and rails.
- **Head-** uppermost horizontal part of frame.
- **Rebate-** The depression made inside the frame to receive the shutter.
- **Sill-** Lowermost horizontal part of window.
- **Style-** Outside vertical members of a shutter are known as styles.

- **Top rail-** The topmost horizontal member of the shutter.
- **Lock rail/ middle rail-** where locking arrangement is fixed.
- **Bottom rail-** The lower most horizontal member of a shutter.
- **Intermediate or cross rail-** Additional horizontal member fixed between the top and bottom rails of shutter.
- **Mullion-** Vertical member running through a shutter.
- **Jamb-** Vertical face of door or window opening which supports the frame.
- **Transom-** Horizontal member of a frame connection two styles of a door or window.







*Fig.6.3: Parts of Window.*

# Types of Door:

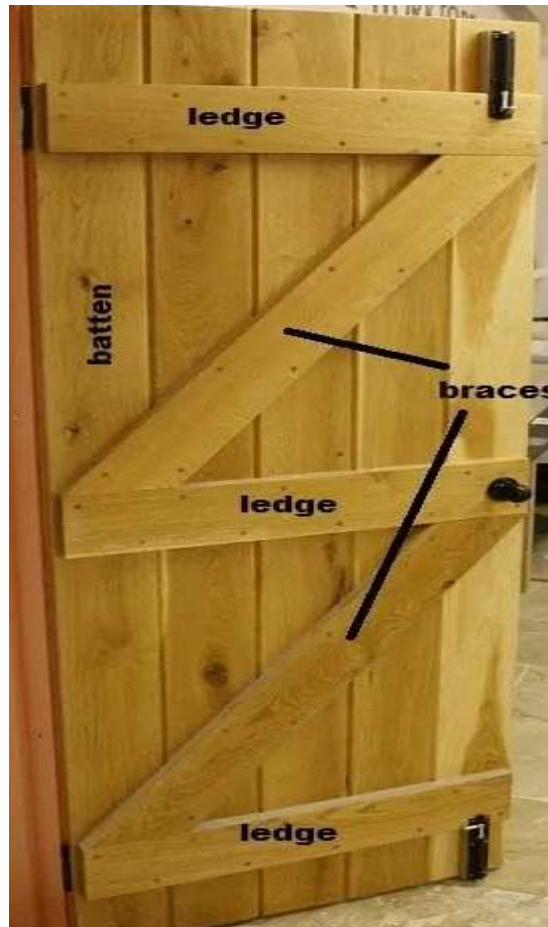
## (a) Battened and Lugged Doors

- Battens are vertical bonds which are having grooves are attached together by horizontal supports called luges.
- Simplest form of door commonly used for narrow openings.
- The door is fixed to the frame by means of T-hinged of iron.



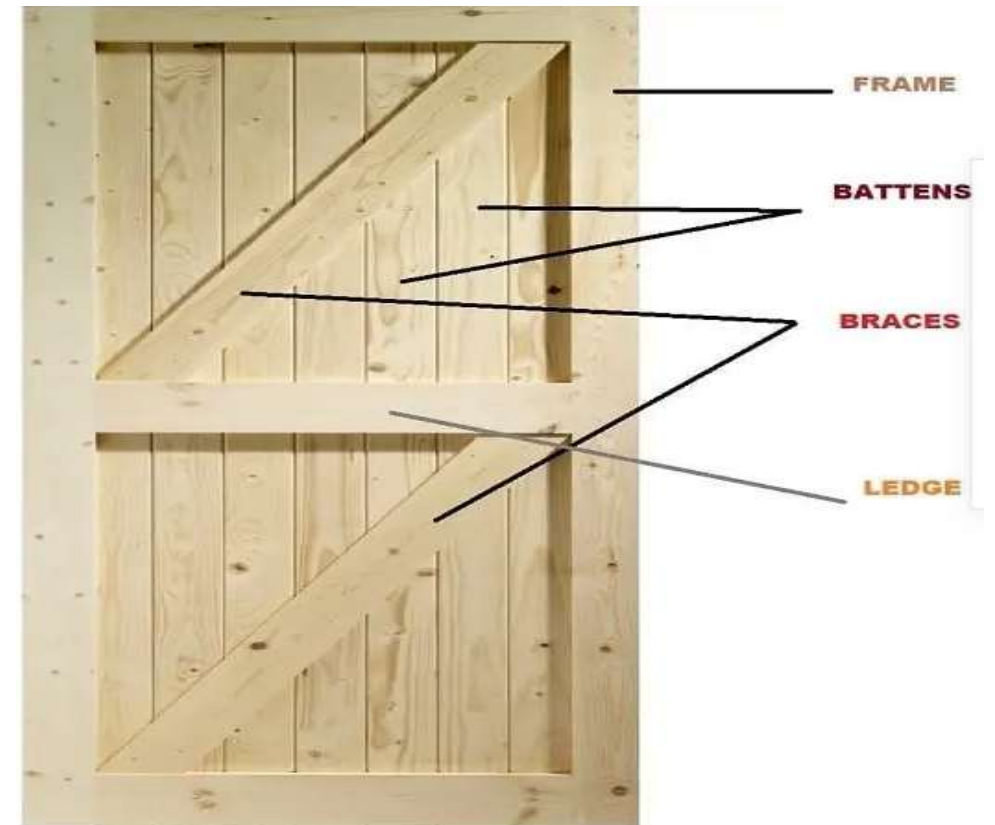
## **b) Battened, Ledged and Braced Doors**

- It is improved form of Battened and Ledged Door and additional inclined members called braces are provided.



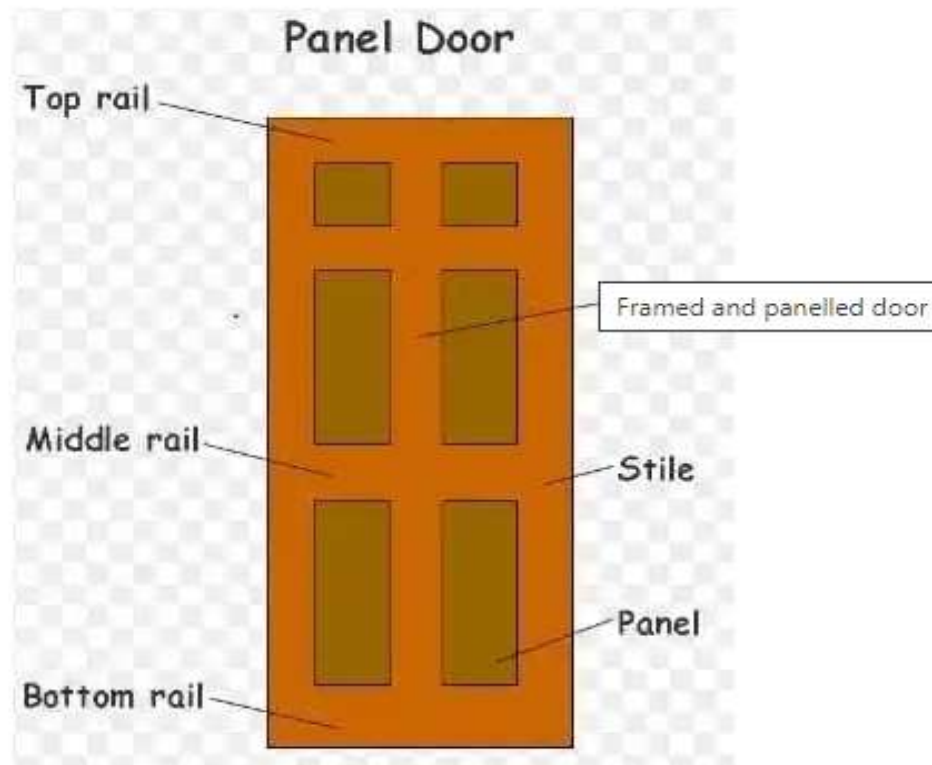
### c) Battened, Ledged, Braced and Framed Doors

- It is improved form of Battened, Ledged and Braced Door.
- The framework of this type of door consist of two vertical members called style and three horizontal members called ledgers or rails, vertical batten and two inclined braces.



## d) Panelled Doors

- These are very strong and will give good appearance when compared to battened doors.
- The panels are made up of timber or plywood or A.C. sheets or glass.



## e) Glazed Doors

- Glazed doors are generally provided in interior wall openings or in hospitals, colleges etc.
- These may be fully glazed or partly glazed and partly panelled.
- Glass panels are provided for glazed doors.



## **f) Flush Doors**

- In flush doors, a solid or semi-solid or core portion is covered on both sides with plywood or face veneer.
- Now a days these type of doors are widely used because of good appearance, economic, ease of construction and greater durability.

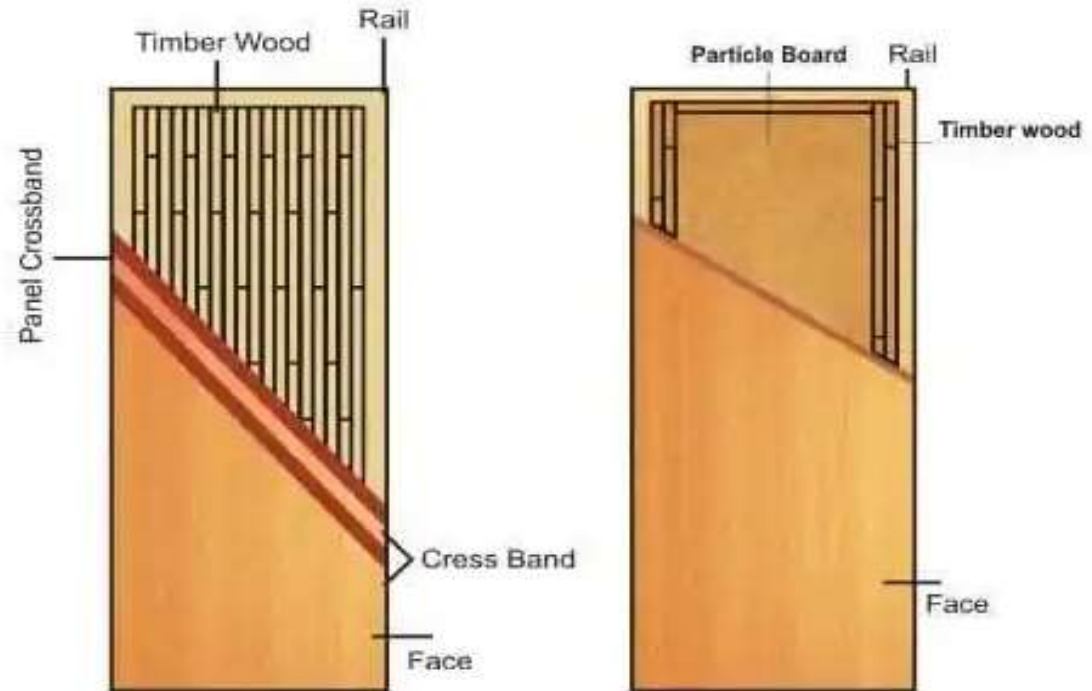
**❖ There are two types of flush door:**

- i) Solid Core or Laminated Core Flush Door
- ii) Cellular core flush door

## i) Solid Core or Laminated Core Flush Door

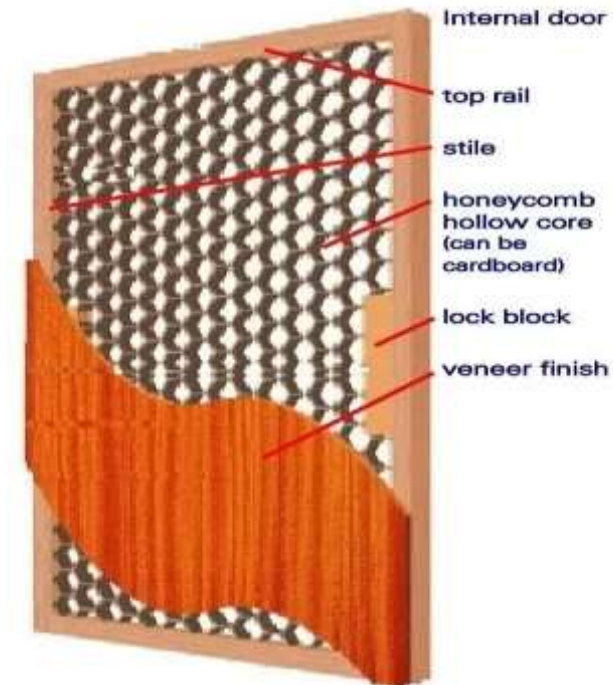
The core part in solid core flush door consists of core strips of timber which are glued under high pressure condition.

Similarly in the laminated core, battens of 25mm width are glued under high pressure.



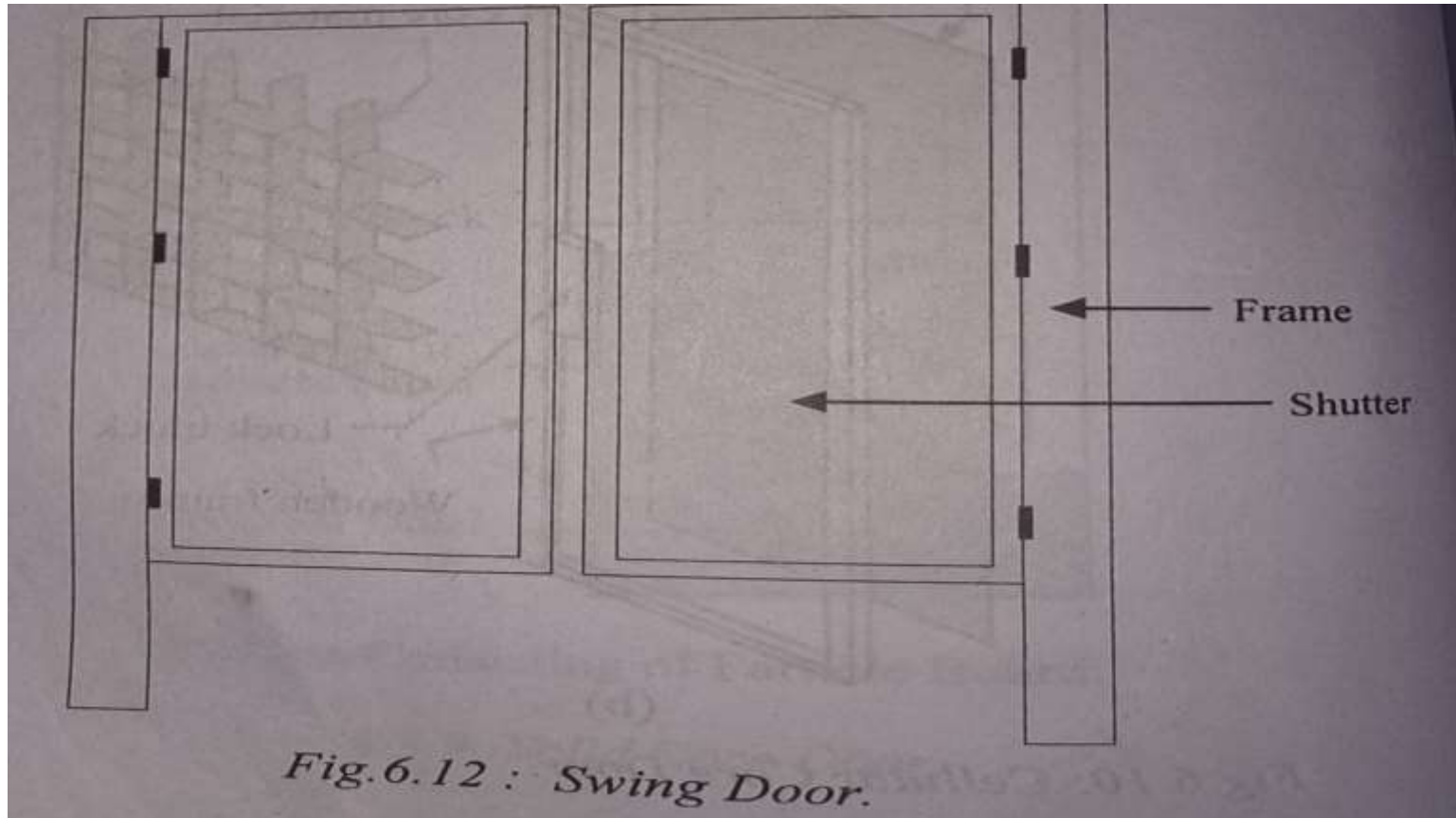
## ii) Hollow core and cellular core flush door

- In this case also stiles and rails are provided for frame. But, a minimum of two intermediate rails should be provided.
- The inner space of door consists of equally space battens of width 25mm each. Other space is called void space which does not exceed 40% of the area of door.



## g) Swing Doors

- In this case, the shutter is attached to frame by double action spring which helps the shutter to move inwards as well as outwards.



## **h) Sliding Doors**

- In this type, with the help of runners and guide rails the door slides to the sides.
- The door may have one or more sliding shutter depending up on the opening available.



## **i)Revolving Doors**

- Revolving doors are only provided in public buildings like museums, banks, libraries etc., because of constant visitors.
- It consists mullion at its center to which four radiating shutters are attached.



## j) Collapsible Steel Doors

- Collapsible steel doors are generally used for workshops, sheds, warehouses etc..
- It acts like a steel curtain which will be opened or closed by horizontal pull or push.



## **k) Rolling Steel Shutter Doors**

- Rolling steel shutter doors are commonly used for warehouses, garages, shops etc..
- These are very strong and offer proper safety to the property. The door consists frame, drum and a shutter of thin steel plate interlocked together.

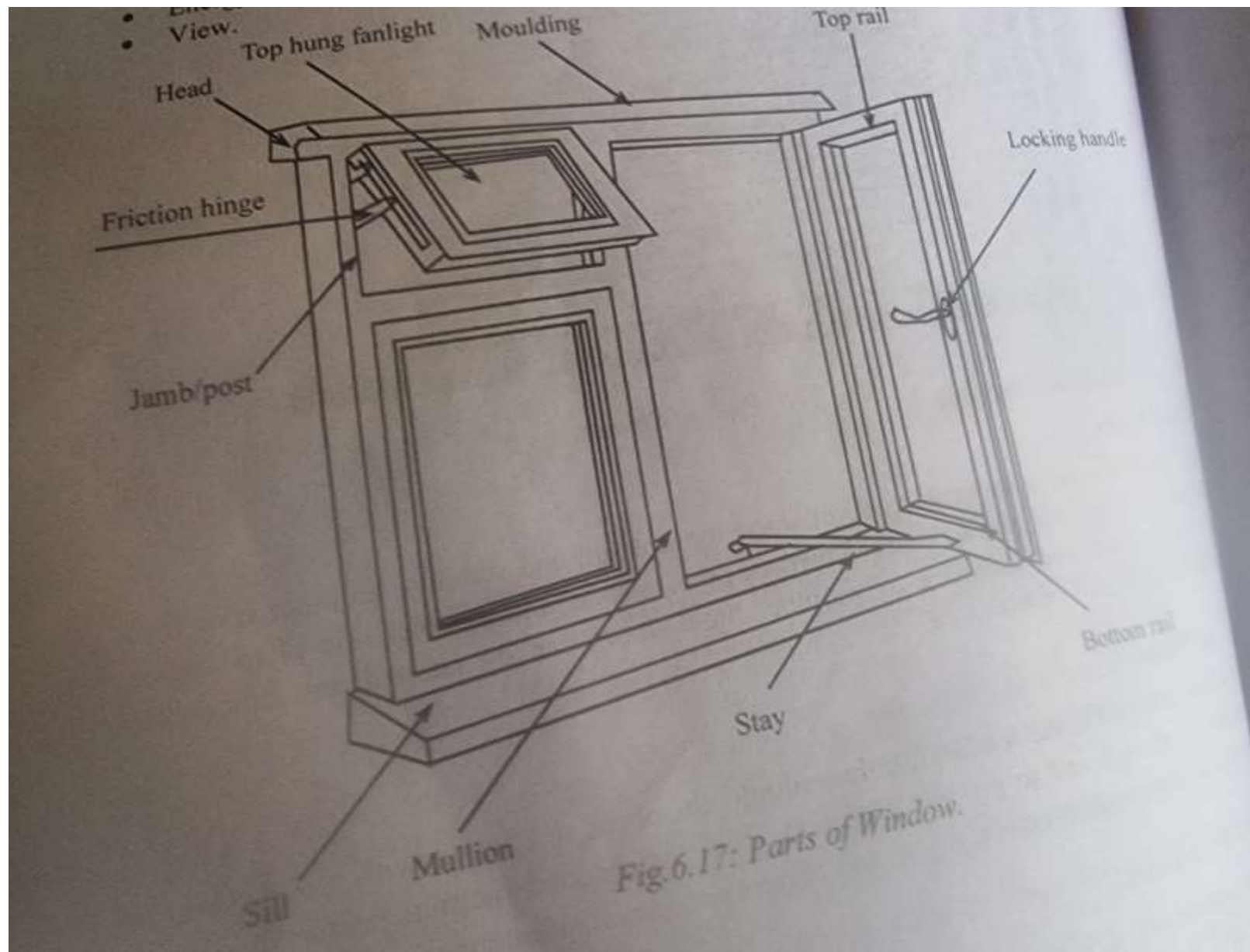


## **Windows And Ventilators:**

- A window is an opening formed in a wall or roof primarily to admit daylight through some transparent or translucent material.
- Windows also serve an important function in providing controlled natural ventilation to buildings and make a major contribution to the visual appearance of buildings.

## **Purpose of window:**

- Admits lights
- Provide fresh air and ventilation
- Beauty
- Energy (heat & cold, in & out)
- View



## **Types of Window:**

### **a) Bay Windows**

- Bay windows are projected windows from wall which are provided to increase the area of opening, which enables more ventilation and light from outside.
- The projection of bay windows are of different shapes.
- It may be triangular or rectangular or polygonal etc.
- They give beautiful appearance to the structure.



## **b)Fixed Windows**

- Fixed windows are fixed to the wall without any closing or opening operation.
- They are provided to transmit the light into the room.
- Fully glazed shutters are fixed to the window frame.
- The shutters provided are generally weatherproof.



### c) Pivoted Windows

- In this type of windows, pivots are provided to window frames.
- Pivot is a shaft which helps to oscillate the shutter.
- No rebates are required for the frame.
- The swinging may either horizontal or vertical based on the position of pivots.



## d) Sliding Windows

- window shutters are movable in the frame.
- The movement may be horizontal or vertical based on our requirements.
- The movement of shutters is done by the provision of roller bearings.
- Generally, this type of window is provided in buses, bank counters, shops, etc..



## e) Casement Windows

- The shutters are attached to frame and these can be opened and closed like door shutters.
- Rebates are provided to the frame to receive the shutters.
- The panels of shutters may be single or multiple.



## **f) Louvered or venetian window**

- A louvered window is one of the best for maximum ventilation.
- This type of window is usually patterned to be sloppy to run water off.



## **g) Dormer Windows**

- Dormer windows are provided for sloped roofs.
- These are projected from the sloping surface as shown in below image.
- They provide ventilation as well as lighting to the room.
- They also enhance aesthetic sense of room.



## **h) Skylight window**

- Skylights are generally provided on the top of sloped roofs.
- To admit light into the rooms, sky lights are provided.
- It is provided parallel to the sloping surface.
- Sky lights can be opened when we required.
- Lead gutters are arranged to frame to make it as waterproof.



## i) Corner Windows

- Corner windows are provided at the corners of room.
- That means corner windows has two faces in perpendicular directions.
- By providing this type of windows, light or air can be entered into room in two different directions.



## **Ventilators:**

- Ventilators are provided for the purpose of ventilation in the room.
- They are provided at greater height than windows nearer to roof level.
- It is in very small size.
- Horizontally pivoted shutters are provided for ventilators.



## **Horizontal and vertical circulation:**

### **❖ Corridors**

- A corridor is a long passage or hallway in a building that provides access to different rooms, spaces, or sections of the building.
- It serves as a circulation area for the movement of people.

### **Functions of Corridors:**

- Provide access to rooms and facilities.
- Facilitate smooth movement of occupants.
- Serve as emergency escape routes during fire or other emergencies.
- Connect different parts of a building.

# **Types of Corridors**

## **1. Single-Loaded Corridor**

- Rooms are located on only one side of the corridor.
- Common in hotels, hospitals, and educational buildings.

## **2. Double-Loaded Corridor**

- Rooms are located on both sides of the corridor.
- More space-efficient and commonly used in apartments, hotels, and office buildings.

### **3. Open Corridor**

- One side of the corridor is open to the outside environment.
- Provides natural light and ventilation.

### **4. Central Corridor**

- Located in the center of the building with rooms on both sides.
- Common in multi-story buildings.

## **Requirements of a Good Corridor**

- Adequate width for safe movement.
- Proper lighting and ventilation.
- Non-slip flooring.
- Clear and unobstructed pathways.
- Compliance with fire safety regulations.

## **Advantages**

- Improves circulation within a building.
- Enhances accessibility.
- Facilitates emergency evacuation.
- Organizes building layout efficiently.

## **Disadvantages**

- Occupies usable floor area.
- May increase construction costs.
- Poorly designed corridors can cause congestion.

# Ladder and Its Types:

## Definition:

- Ladders are ancient tools and technology.
- Ladder is a vertical or inclined set of rungs or steps used for climbing or descending.
- It is commonly used in construction, maintenance, and household work.



## **1. Straight Ladder (Single Ladder)**

- Consists of a single section.
- Fixed length and cannot be adjusted.
- Used for simple access to elevated places.

## **2. Extension Ladder**

- Made of two or more sections that slide over one another.
- Length can be adjusted as needed.
- Commonly used for reaching high locations.

### **3. Step Ladder**

- Self-supporting ladder with an A-shaped frame.
- Does not require wall support.
- Commonly used indoors for maintenance work.

### **4. Folding Ladder**

- Can be folded for easy storage and transportation.
- Suitable for household and light construction work.

## **5. Platform Ladder**

- Has a working platform at the top.
- Provides better stability and safety.
- Used for electrical and maintenance work.

## **6. Trestle Ladder**

- Consists of two ladders joined by a hinged support.
- Used by painters and decorators for indoor work.

## **7. Fixed Ladder**

- Permanently attached to a structure.
- Used in water tanks, silos, towers, and industrial buildings.

## **8. Rope Ladder**

- Made of ropes with wooden or metal rungs.
- Lightweight and portable.
- Used in emergency situations and marine applications.

## **Characteristics of a Good Ladder**

- Strong and durable.
- Lightweight and easy to handle.
- Slip-resistant rungs.
- Proper spacing between rungs.
- Safe and stable during use.

## **Staircase:**

- A staircase is a structural element that is used for travelling to a higher height or it is used to travel from one storey of a building to another.
- It is designed to cover a large vertical distance by cleaving up into smaller vertical distances which are called steps.
- The steps may be moving continuously that facilitates the reaching of heights without climbing.
- A staircase is access between the floors of building.

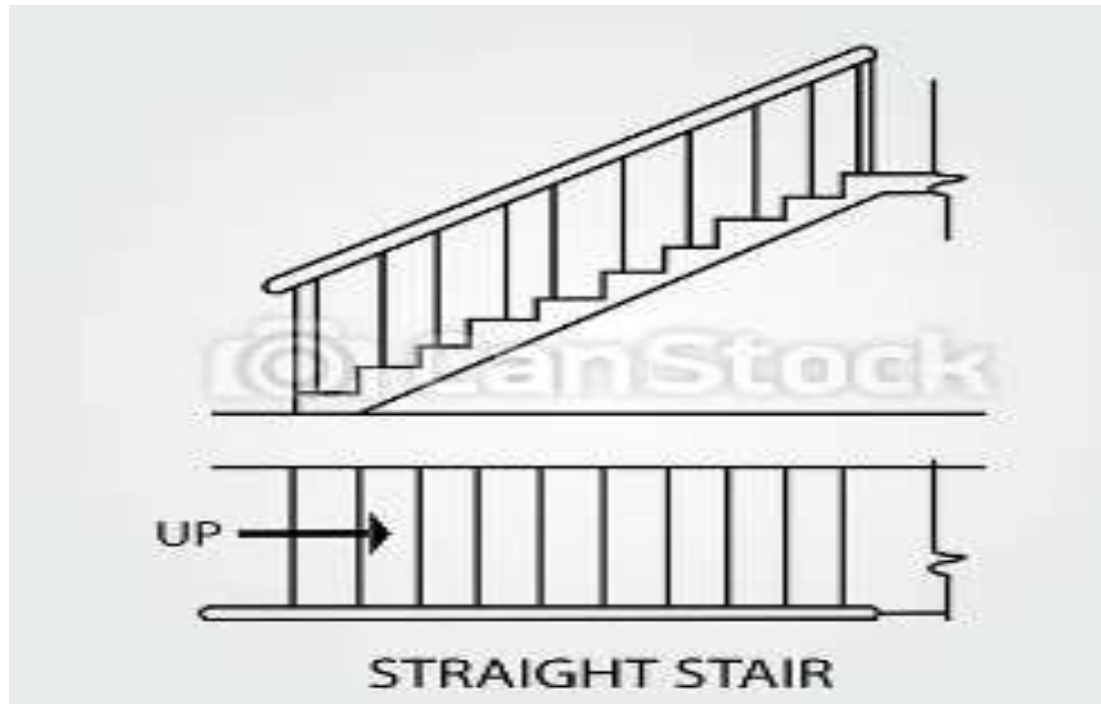
## **Function of Staircase:**

- Provide an access from one floor to another.
- Provide a safe means of travel between floors.
- Provide a suitable means of escape in case of fire.
- Provide a mean of conveying fittings and furniture between floor levels.

# Types of Stairs:

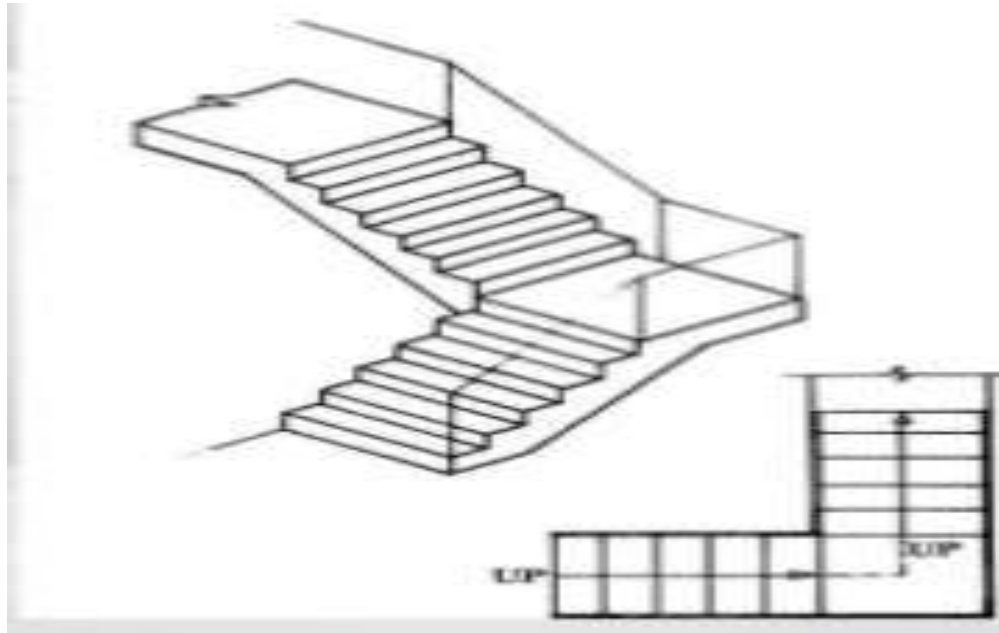
## 1. Straight Stair:

- Stair may have one flight or two flights with a landing.
- Only for small buildings/domestic buildings.



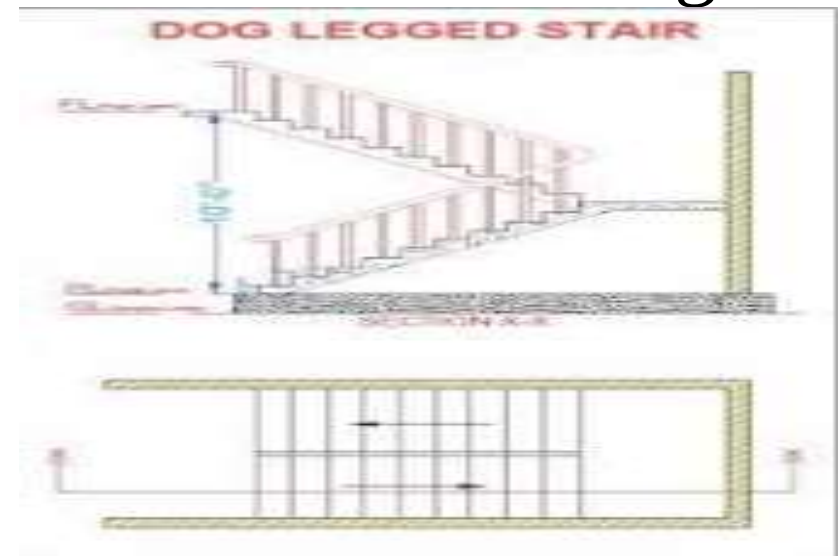
## 2. Quarter Turn Stair:

- A stair turning through one right angle is known as quarter turn stairs.
- Rises to a landing between two floors, turns through 90 degree, then rises to the floor above.



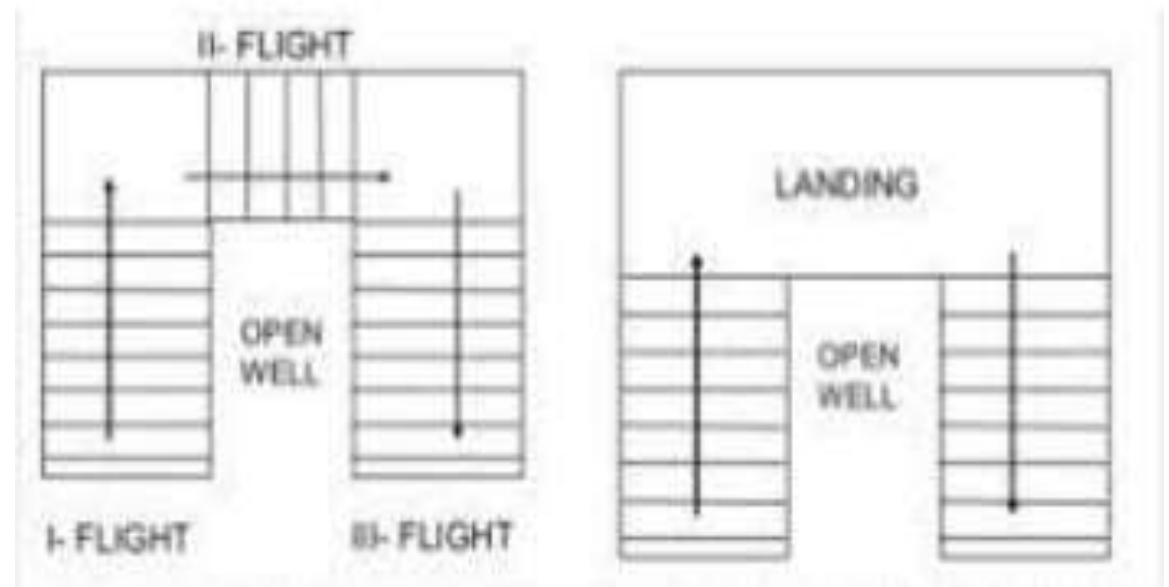
### 3. Dog Legged/ Half turn Stair:

- Newel post are provided at the beginning and end of each flight.
- There is no space between two flights.
- A half spaced landing is provided generally to affect the change in direction.
- Two straight flights at 180 degree turn with mid landing.
- Used in most of the building.



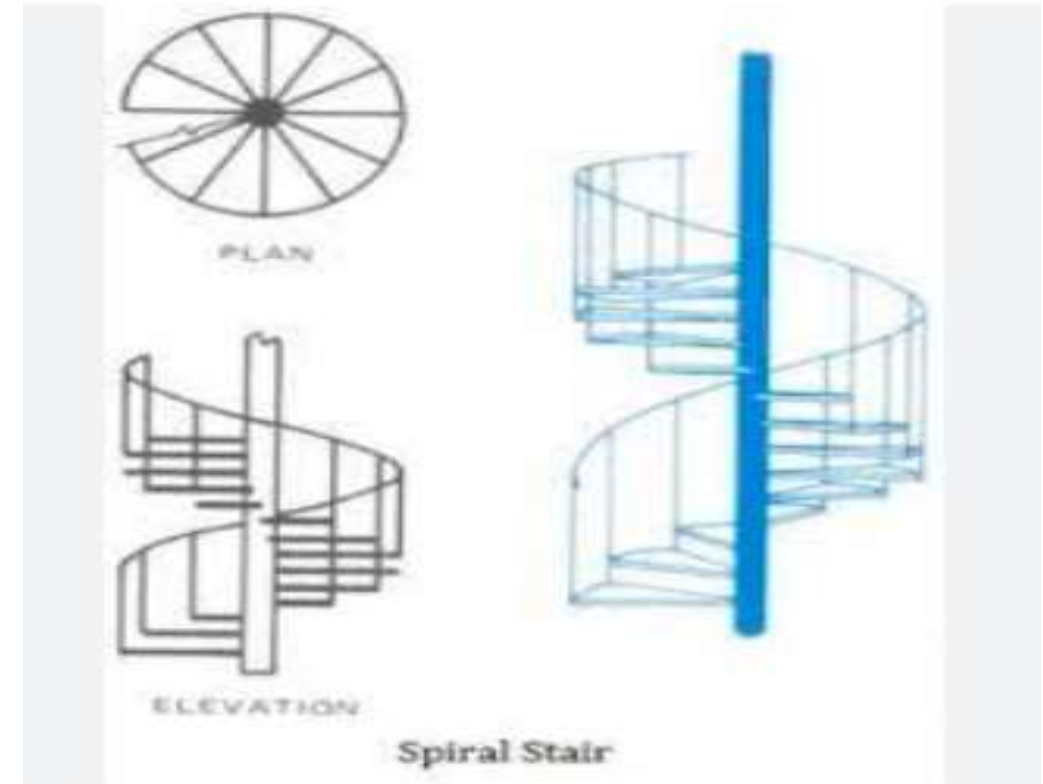
#### 4. Open Well/ Open Newel Stair:

- In these type of stair there is a well or opening between the flights in plan.
- This well may be rectangular or of any geometrical shape.
- Two or more than two straight flights arranged around a well or space.
- Used in public building



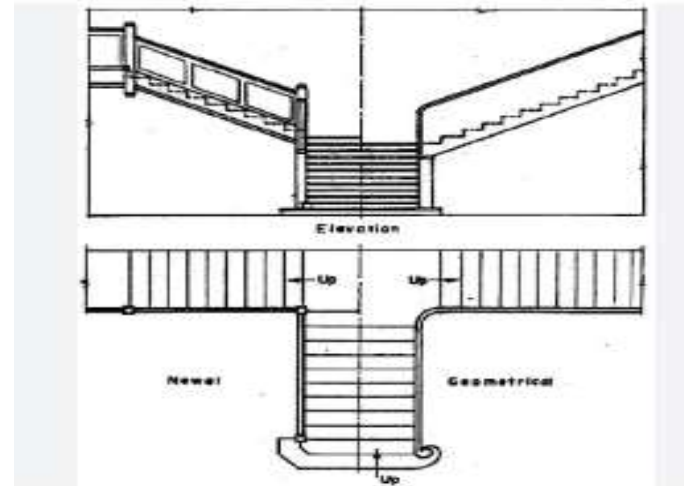
## 5. Spiral staircase:

- A spiral staircase is a structure that is built around a central post that acts as an axis for its steps.
- Suitable where little space is available.
- Economical.
- Made from steel and welded together.



## 6. Bifurcated Stairs:

- This type of stair is commonly used in public buildings at their entrance hall.
- The stair has a wider flight at the bottom, which bifurcates into two narrower flights, one turning to the left and the other to the right, at the landing.
- It may be either of newel type with a newel post .



## Ramps:

- A sloping surface connecting two levels.
- It is another means of ascent & descent between the landing or floors of a building.



## Lift:

- It is a lifting machine or appliance having a platform.
- To move or bring (something) upward from the ground or other support to a higher position.



## Escalator:

- An escalator is a moving staircase which carries people between floors of a building or structure.
- It consists of a motor-driven chain of individually.



## Elevator:

- An elevator or lift is a machine that vertically transports people or freight between levels.
- They are typically powered by electric motors that drive traction .



# **Flooring and Flooring types:**

## **Definition:**

- Flooring is the finished upper surface of a floor that provides a smooth, durable, and comfortable walking surface.
- It enhances the appearance, strength, and functionality of a building.

## **Requirements of Good Flooring:**

- Durable and long-lasting, Hard and wear-resistant
- Easy to clean and maintain
- Economical
- Non-slippery
- Good appearance

## **Types of Flooring:**

### **1. Cement Concrete Flooring**

- Made from cement, sand, and aggregate.
- Economical and durable.
- Commonly used in houses, garages, and warehouses.



**Fig: Cement Concrete Flooring**

## 2. Terrazzo Flooring

- Made of marble chips embedded in cement.
- Attractive, durable, and easy to maintain.
- Used in schools, hospitals, and public buildings.



### 3. Tile Flooring

- Made of ceramic, vitrified, porcelain, or mosaic tiles.
- Available in various colors and designs.
- Commonly used in residential and commercial buildings.



## 4. Marble Flooring

- Made from natural marble stone.
- Elegant appearance and high durability.
- Used in luxury homes, hotels, and offices.



Fig: Marble Flooring

## 5. Granite Flooring

- Made from natural granite stone.
- Very strong, durable, and resistant to wear.
- Suitable for high-traffic areas.



## 6. Wooden Flooring

- Made from timber or engineered wood.
- Provides warmth and aesthetic appeal.
- Used in homes, halls, and offices.



## 7. Stone Flooring

- Made from natural stones such as slate, limestone, or sandstone.
- Durable and attractive.
- Suitable for both indoor and outdoor areas.



Fig: Flagstones Flooring Material

## 8. Brick Flooring

- Constructed using bricks laid on a prepared base.
- Economical and durable.
- Used in courtyards, warehouses, and rural buildings.



Fig: Brick Flooring in Buildings

## 9) Mud and Murram

- Mud is nothing but moist earth which is being used as flooring material in since olden days.
- Murram is also direct nature product which is obtained from the disintegration of rock.



Fig: Mud Flooring



Fig: Murram as a Flooring Material

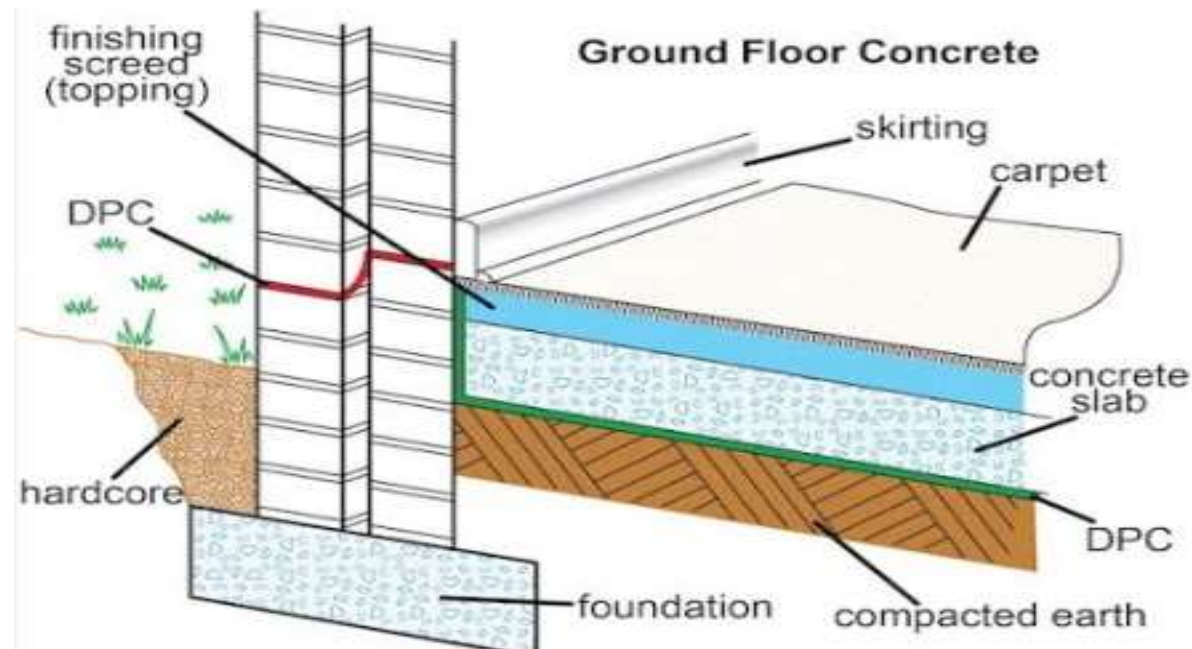
## 10) Plastic or P.V.C. flooring

- Made of plastic material called poly vinyl chloride(P.V.C). Fabricated in the form of tiles of different size & color.
- Costly & slippery and can be damaged very easily when it comes in the contact with burning objects.



## Solid ground floor:

- No gap between G/L & plinth level.
- The gap between G/L & plinth level are completely filled with solid materials.
- A solid ground floor consists of a layer of concrete.



## Suspended ground floor:

- A suspended floor is a ground floor with a void underneath the structure.
- The floor can be formed in various ways, using timber joists, precast concrete panels, block and beam system or cast in-situ with reinforced concrete.
- Floor that is not touching the ground and suspended above the ground.



## **Building services:**

### **❖ Sanitary fittings:**

#### **Definition**

- Sanitary fittings are fixtures and appliances installed in a building for the collection, use, and disposal of water and waste.
- They ensure hygiene, cleanliness, and comfort for occupants.

# **Common Sanitary Fittings**

## **1. Water Closet (WC)**

- Commonly known as a toilet.
- Used for the disposal of human waste.

## **2. Wash Basin**

- Used for washing hands, face, and small items.
- Installed in bathrooms, kitchens, and washrooms.

## **3. Sink**

- Used for washing utensils and food items.
- Commonly installed in kitchens and laboratories.

## **4. Urinal**

- Sanitary fixture used for urination.
- Commonly found in public toilets and commercial buildings.

## **5. Bathtub**

- Used for bathing.
- Available in different sizes and materials.

## **6. Shower**

- Provides water for bathing through a shower head.
- Commonly installed in modern bathrooms.

## **Requirements of Good Sanitary Fittings**

- Durable and corrosion-resistant.
- Smooth and easy to clean.
- Leak-proof and hygienic.
- Economical and easy to maintain.
- Attractive appearance.

## **Importance of Sanitary Fittings**

- Maintain hygiene and public health.
- Ensure proper water supply and waste disposal.
- Improve comfort and convenience.
- Prevent water leakage and contamination.

# **Rain Water Harvesting:**

## **Definition**

- Rain water harvesting is the process of collecting, storing, and using rainwater that falls on rooftops, land surfaces, or catchment areas for future use.
- It is an important method of water conservation.

# **Methods of Rain Water Harvesting**

## **1. Rooftop Rain Water Harvesting**

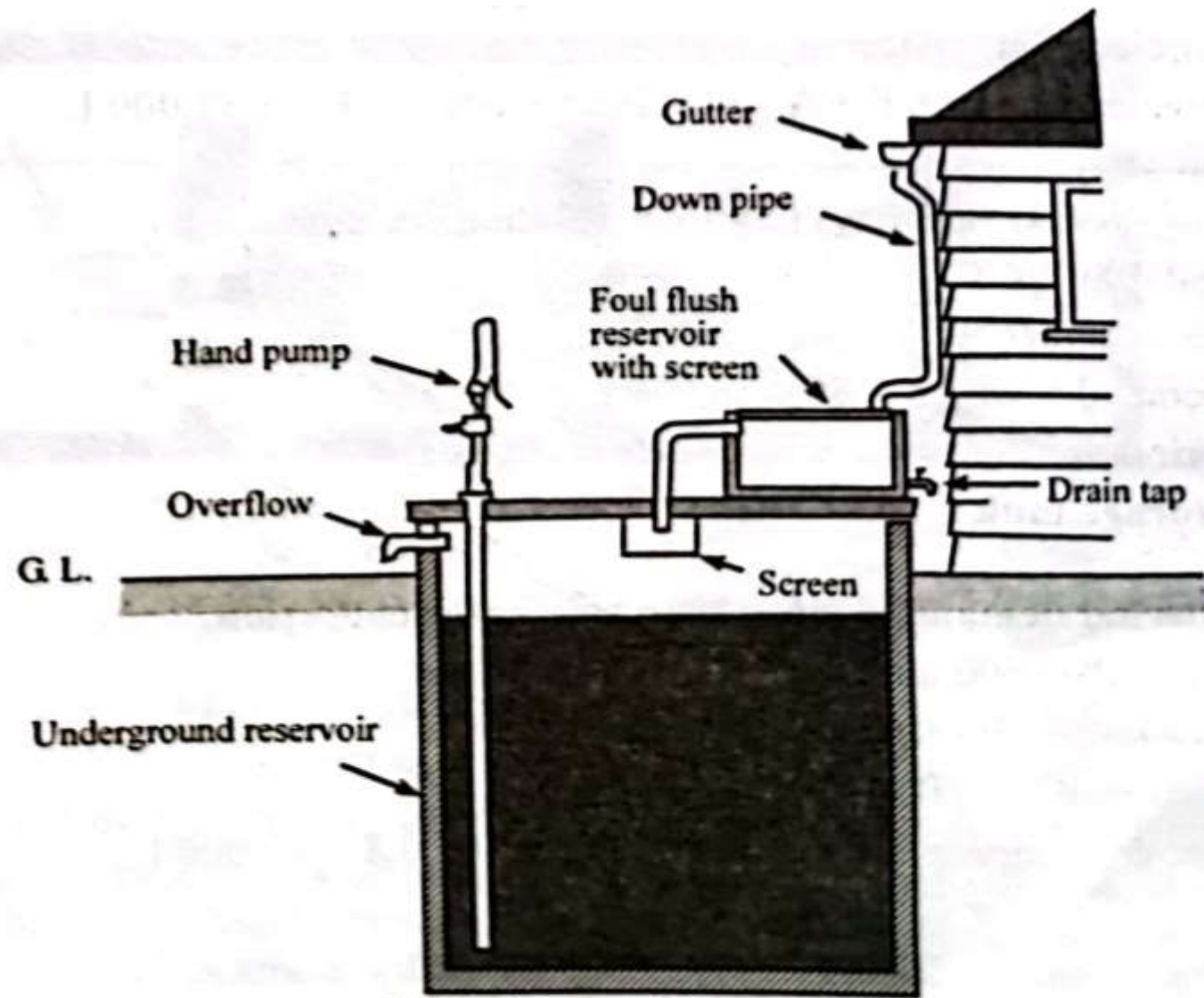
- Rainwater is collected from roofs of buildings.
- Water is directed through pipes into storage tanks or recharge pits.
- Most common method in residential buildings.

## **2. Surface Runoff Harvesting**

- Rainwater flowing over land surfaces is collected.
- Stored in ponds, tanks, or underground reservoirs.
- Useful in parks, fields, and open areas.

## **Components of Rain Water Harvesting System:**

- Catchment area (roof or ground surface)
- Gutters and downpipes
- Filter unit (removes dust and impurities)
- Storage tank or reservoir
- Recharge pit / well
- Overflow system



(b) Underground Reservoir

## **Advantages**

- Reduces water shortage problems
- Recharges groundwater level
- Improves water availability in dry seasons

## **Disadvantages**

- Requires initial installation cost
- Needs regular cleaning and maintenance
- Storage capacity is limited

# **Wiring Systems (Building Electrical Services):**

## **Definition:**

- A wiring system is the arrangement of electrical conductors, switches, sockets, and protective devices used to distribute electricity safely and efficiently throughout a building.

## **Objectives of Wiring Systems**

- Supply electricity to different parts of a building
- Ensure safety of users and equipment
- Provide control over electrical appliances
- Reduce energy loss and hazards like short circuit or fire

# **Types of Wiring Systems:**

## **1. Cleat Wiring**

- Wires are supported on porcelain or plastic cleats.
- Simple and cheap installation.
- Temporary wiring system.

## **2. Casing and Capping Wiring**

- Wires are placed inside wooden or PVC casing and covered with a cap.
- Provides protection against mechanical damage.
- Common in residential buildings (older system).

### **3. Batten Wiring (TRS/CTS Wiring)**

- Wires are fixed on wooden battens using clips.
- Simple and economical.
- Suitable for dry locations.

### **4. Conduit Wiring**

- Wires are carried through metal or PVC pipes (conduits).
- Highly safe and durable.
- Most commonly used in modern buildings.

# **Components Used in Wiring Systems**

- Wires and cables
- Switches and sockets
- Distribution board (DB)
- Circuit breakers and fuses
- Junction boxes
- Earthing system

## **Requirements of Good Wiring System**

- Safe and shock-proof
- Durable and long-lasting
- Easy to maintain and repair
- Economical
- Properly insulated
- Fire-resistant

# **Lightning Arrester:**

## **Definition**

- A lightning arrester is a protective device used in buildings and electrical systems to protect structures and equipment from damage caused by lightning strikes.
- It safely conducts the high-voltage surge from lightning into the earth.

## **Working Principle:**

- When lightning strikes a building or nearby power line, a high-voltage surge is produced.
- The lightning arrester provides a low-resistance path to ground.
- The electrical surge is diverted safely into the earth through the earthing system.
- This prevents damage to electrical appliances and the building structure.

## **Requirements of a Good Lightning Arrester:**

- Low resistance earthing system
- Strong and corrosion-resistant materials
- Proper height above the structure
- Efficient discharge capacity
- Proper installation and maintenance

## **Advantages:**

- Protects buildings from fire and structural damage
- Safeguards electrical appliances
- Improves safety of occupants
- Reduces risk in power systems

**Thank You**